

Invertibility of matrices via REF/RREF

$A \in M_{n \times n}$ is invertible iff a REF of A has n leading entries

Check through RREF

$$(A | I_n) \xrightarrow{\text{RREF}} (I_n | A^{-1})$$

Change of Coordinates

How are $[v]_{\beta}, [v]_{\beta'}$ related? β, β' finite bases.

$$[v]_{\beta} = [I_V(v)]_{\beta'} = [I_V]_{\beta}^{\beta'} [v]_{\beta}$$

Inverse of change of coordinate matrix.

$$\text{Thm } ([I_V]_{\beta}^{\beta'})^{-1} = [I_V]_{\beta'}^{\beta}$$

$$\text{Pf } [I_V]_{\beta}^{\beta'} [I_V]_{\beta'}^{\beta} = [I_V \circ I_V]_{\beta'}^{\beta'} = [I_V]_{\beta'}^{\beta'} = I$$

Change of bases

Given $T: V \rightarrow W$, finite bases β, β' for V , finite bases γ, γ' for W ,

how are $[T]_{\beta}^{\gamma}$ and $[T]_{\beta'}^{\gamma'}$ related?

$$\begin{aligned} [T]_{\beta'}^{\gamma'} &= [I_W \circ T \circ I_V]_{\beta'}^{\gamma'} \\ &= [I_W]_{\gamma'}^{\gamma'} [T \circ I_V]_{\beta'}^{\gamma'} \\ &= [I_W]_{\gamma'}^{\gamma'} [T]_{\beta}^{\gamma} [I_V]_{\beta'}^{\beta} \end{aligned}$$

Similar matrices

Def $A, B \in M_{n \times n}$ are similar if $\exists Q \in M_{n \times n}$ such that $B = Q^{-1} A Q$.

$$\text{Let } L_A^k = \underbrace{L_A \circ L_A \circ L_A \cdots \circ L_A}_{k \text{ times}}$$

Matrix representation of orthogonal projection

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the orthogonal projection onto W .

$$T(w_1) = w_1, \quad T(w_2) = w_2.$$

Find $\beta' = \{w_1, w_2, w_3\}$ such that $w_1 = w_1 \times w_2 = (1, 1, 1)$

$$\text{and } T(w_3) = (0, 0, 0)$$